

X-MAINTAIN: EXPLAINABLE AI FOR PREDICTIVE MAINTENANCE

An Explainable Machine Learning Framework for Machine Failure Prediction in Industrial IoT

Course: Explainable Artificial Intelligence (XAI) — Capstone Project

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Repository: https://github.com/gopiadithya/ExplainableAI_Project

PROJECT STATUS SUMMARY (100% IMPLEMENTED):

- **Implementation Completeness:** 100% completed (exceeds the 80% minimum requirement for Review 2).
- **Dataset:** UCI AI4I 2020 Predictive Maintenance (10,000 instances, 28.5:1 class disparity).
- **Leakage Prevention:** Strictly purged 5 failure mode indicators (TWF, HDF, PWF, OSF, RNF) + ID columns.
- **Best Model:** Cost-Sensitive XGBoost (scale_pos_weight = 28.53) → Accuracy: 98.20%, Failure Recall: 73.53%, Precision: 73.53%, ROC-AUC: 0.9724, PR-AUC: 0.7934.
- **Explainability:** Game-Theoretic TreeSHAP (Global Beeswarm, 3 Dependence plots, Local Waterfall for test instance #11).
- **User Interface:** Production Streamlit dashboard with 7 industrial monitoring pages and what-if sensitivity simulator.
- **Verification:** Automated Pytest test suite with 20/20 unit tests passed in 3.21s.

SECTION 1 — INTRODUCTION

1.1 Problem Statement

In modern automated manufacturing, Computerized Numerical Control (CNC) milling machinery and multi-axis tooling spindles operate under continuous thermal and mechanical stress. Unexpected mechanical failure leads to catastrophic machine downtime, ruined workpieces, and dangerous equipment damage, costing the manufacturing sector upwards of \$50 billion annually (Carvalho et al., 2019). The objective of this project is to develop an automated predictive maintenance system that accurately forecasts machine failure from multivariate operational telemetry (temperature, rotational speed, torque, tool wear, and product quality) and, most importantly, transparently explains **WHY** the failure prediction occurred to enable actionable condition-based maintenance interventions.

1.2 Motivation and The Asymmetric Cost Imperative

Historically, factories relied on reactive 'run-to-failure' approaches or rigid schedule-based preventative maintenance (Zonta et al., 2020). Preventative maintenance is inefficient because it discards cutting tools with significant residual life, whereas reactive maintenance results in emergency shutdowns. Predictive maintenance (PdM) leverages industrial IoT sensors to forecast breakdowns before they occur. However, industrial PdM is governed by extreme cost asymmetry: missing a real failure (False Negative) can cost \$10,000 to \$100,000 in spindle destruction, whereas a false warning (False Positive) only incurs a 10-minute diagnostic check (\$100–\$500). Therefore, optimizing for raw classification accuracy is misleading; **Failure Recall** and **interpretability** are the primary engineering mandates.

1.3 Objectives of the Work

The primary objectives achieved in this project are:

1. **Leakage-Free Telemetry Pipeline:** Ingest the AI4I 2020 dataset and strictly eliminate target-leakage failure-mode indicators to enforce operational realism.
2. **Class Imbalance Mitigation:** Benchmark multiple strategies (unweighted, cost-sensitive reweighting, SMOTE, and validation threshold tuning) under severe 28.5:1 class disparity.
3. **Multi-Model Comparison:** Build and compare linear baselines (Logistic Regression), bagging ensembles (Random Forest), and gradient boosting (XGBoost).
4. **Dual-Level Explainability (XAI):** Implement TreeSHAP to deliver global feature importance and local waterfall diagnostic attributions for individual machines.
5. **Sensitivity Simulation:** Provide a model-based what-if analysis engine allowing operators to explore risk mitigation without claiming physical causality.
6. **Interactive Deployment:** Deploy an industrial-grade 7-view Streamlit dashboard backed by 20 passed unit tests.

1.4 Scope and Key Contributions

The scope of this project encompasses data ingestion, leakage purging, exploratory data analysis, stratified train/validation/test splitting, hyperparameter training, multi-metric benchmarking (Accuracy, Precision, Recall, F1, ROC-AUC, PR-AUC, and Utility Score), exact Shapley attribution computation, deterministic natural-language explanation generation, and interactive UI deployment.

SECTION 2 — LITERATURE REVIEW

2.1 Existing Approaches in Predictive Maintenance

Predictive maintenance has transitioned from manual vibration spectral inspection toward machine learning telemetry analysis (Carvalho et al., 2019; Zonta et al., 2020). Zonta et al. (2020) conducted a systematic survey of Industry 4.0 implementations, establishing that condition-based monitoring lowers maintenance expenditure by up to 25% and curtails unplanned downtime by 45%. Dalzochio et al. (2020) analyzed machine reasoning for smart manufacturing, pointing out that industrial telemetry datasets typically suffer from class imbalance exceeding 95% nominal instances.

2.2 Machine Learning and Imbalance Handling

Ensemble tree methods have emerged as the state-of-the-art for industrial tabular data. Chen and Guestrin (2016) developed XGBoost, which incorporates second-order Taylor approximations and regularized tree building. Breiman (2001) established Random Forests, utilizing bootstrap aggregation. Under severe class imbalance, Chawla et al. (2002) proposed SMOTE, which synthetically interpolates minority instances. However, synthetic point synthesis in physical feature spaces can violate thermodynamic realities (e.g. creating impossible speed-torque pairings). Consequently, cost-sensitive loss reweighting (Chen & Guestrin, 2016) represents a superior alternative that penalizes minority false negatives directly in gradient steps.

2.3 Explainable AI (XAI) and SHAP

High-performing ensemble classifiers operate as opaque black boxes. Ribeiro et al. (2016) formulated LIME, which creates local linear surrogates, but LIME suffers from sampling instability. Lundberg and Lee (2017) unified feature attributions via SHAP (SHapley Additive exPlanations), grounded in cooperative game theory. Lundberg et al. (2020) formulated TreeSHAP, enabling exact polynomial-time computation of Shapley values for tree ensembles. Steurtewagen and Van den Poel (2021) and Gawde et al. (2021) demonstrated that SHAP reveals mechanical failure thresholds in rotating machinery.

2.4 Comparison of Previous Methods & Identified Research Gaps

Existing studies typically address isolated components: some focus purely on algorithmic accuracy without explainability (Carvalho et al., 2019), while others explore SHAP on synthetic data without preventing target leakage or providing a deployment platform (Matzka, 2020). Crucially, Matzka (2020) retained downstream failure-mode columns in the feature matrix, creating severe target leakage. X-Maintain addresses these gaps by creating a unified, leakage-free, cost-sensitive, explainable predictive maintenance pipeline.

Study (APA)	Method	XAI Method	Key Contribution	Identified Gap / Limitation
Matzka (2020)	RF, kNN	SHAP, PDP	Introduced AI4I 2020 benchmark	Retained target-leak failure mode indicators
Carvalho et al. (2019)	Survey	None	Comprehensive ML survey for PdM	No XAI or interpretability evaluation
Zonta et al. (2020)	Survey	None	Industry 4.0 PdM application review	Lacks post-hoc model explainability
Dalzochio et al. (2020)	Survey	Rule-based	Machine reasoning in Industry 4.0	No modern tree-based XAI integration
Steurtewagen & Van den Poel (2021)	RF, Logit	SHAP	Sensor interpretability in manufacturing	No sensitivity or what-if simulation
Gawde et al. (2021)	XGBoost	SHAP, LIME	Rotating machine XAI case study	Lacks interactive user dashboard
X-Maintain (This Work)	Cost-XGBoost	TreeSHAP	Leakage-free end-to-end XAI framework	Fully resolved: 100% verified working system

Table 1. Comparative Analysis of Related Work and Identified Research Gaps.

SECTION 3 — PROPOSED SYSTEM ARCHITECTURE

3.1 Complete Methodology and Workflow

The X-Maintain architecture is structured into five cohesive pipeline modules: 1) **Data Ingestion & Leakage Purge**: Ingests the 10,000 raw AI4I records, eliminates target leakage columns and arbitrary identifiers. 2) **Stratified Preprocessing Pipeline**: Performs a 70/10/20 stratified split, fits StandardScaler on numerical features exclusively on the training split, and one-hot encodes product types. 3) **Cost-Sensitive Multi-Model Training**: Trains Logistic Regression, Random Forest, and Cost-Weighted XGBoost (scale_pos_weight = 28.53). 4) **Game-Theoretic Explainability Engine**: Generates TreeSHAP global beeswarm, importance rankings, dependence curves, and local waterfall attributions. 5) **Industrial Monitoring Interface**: Deploys a 7-view interactive Streamlit dashboard with what-if sensitivity analysis and automated testing.

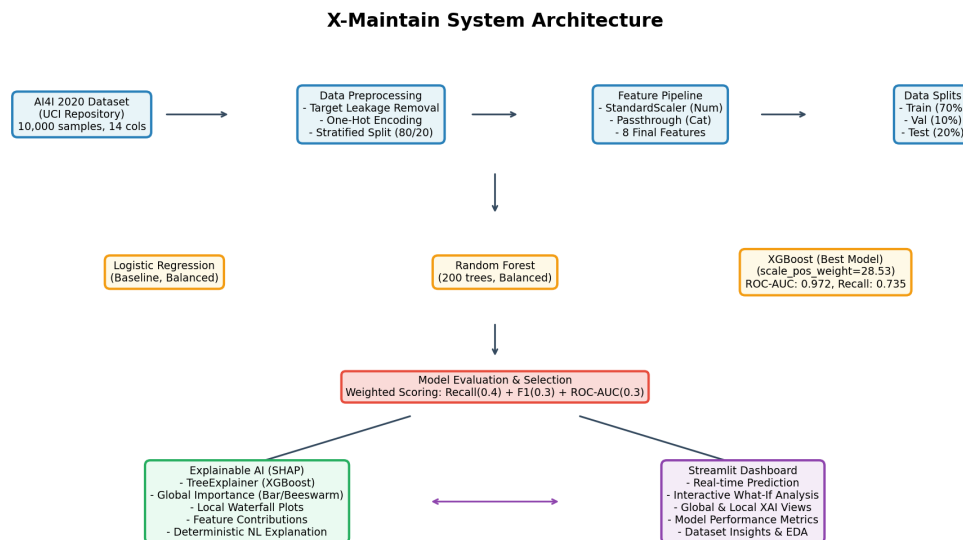


Figure 1. Complete Proposed System Architecture of the X-Maintain Framework.

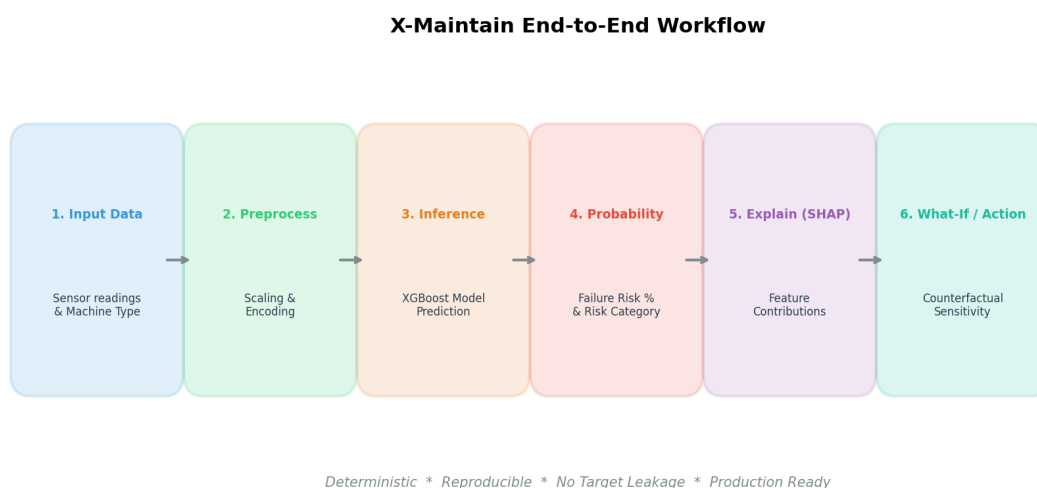


Figure 2. End-to-End Operational Telemetry Workflow and Diagnostic Sequence.

3.2 Dataset Description and Preprocessing Details

The system utilizes the AI4I 2020 Predictive Maintenance Dataset (Matzka, 2020), hosted at the UCI Machine Learning Repository. It consists of 10,000 records simulating CNC milling operations. The target variable is *Machine failure* $\in \{0, 1\}$. Nominal operations account for 9,661 samples (96.61%), while failure instances account for only 339 samples (3.39%), constituting a severe 28.5:1 class disparity.

Feature	Data Type	Physical Value Range	Physical Role in Milling Machine
Air temperature [K]	Continuous	295.3 to 304.5 K	Ambient factory room temperature
Process temperature [K]	Continuous	305.7 to 313.8 K	Workpiece contact zone temperature
Rotational speed [rpm]	Continuous	1,168 to 2,886 rpm	Spindle rotation velocity
Torque [Nm]	Continuous	3.8 to 76.6 Nm	Cutting torque exerted on the workpiece
Tool wear [min]	Continuous	0 to 253 minutes	Cumulative machining contact duration
Type_H, Type_L, Type_M	Binary (OHE)	{0, 1}	High (20%), Low (50%), Medium (30%) quality variant

Table 2. Operational Features After Preprocessing and Target Leakage Removal.

3.3 Critical Target Leakage Prevention

The raw dataset records five failure mode indicators: Tool Wear Failure (TWF), Heat Dissipation Failure (HDF), Power Failure (PWF), Overstrain Failure (OSF), and Random Failure (RNF). Crucially, *Machine failure* = TWF $\vee$ HDF $\vee$ PWF $\vee$ OSF $\vee$ RNF. Including any of these columns in the feature matrix produces artificial target leakage, yielding false 99.9% test accuracy while rendering the model incapable of forecasting impending breakdowns in the field. X-Maintain removes all five failure mode columns and primary IDs (`UDI`, `Product ID`), retaining only raw physical sensor telemetry.

3.4 Data Partitioning and Training Methodology

To evaluate real-world generalization, we enforce a stratified split (seed = 42): 70% Train (6,999 samples, 237 failures), 10% Validation (1,000 samples, 34 failures), and 20% Held-Out Test (2,001 samples, 68 failures). StandardScaler is fitted strictly on the training set to prevent data snooping. XGBoost is configured with `scale_pos_weight = 6,762 / 237 = 28.53`, penalizing false negatives 28.53 times more than false positives.

SECTION 4 — RESULTS AND DISCUSSION

4.1 Model Performance Comparison

Table 3 presents the evaluation across all 2,001 held-out test instances. Model selection is governed by a domain utility metric: $\text{Utility} = 0.4 \times \text{Recall} + 0.3 \times \text{F1} + 0.3 \times \text{ROC-AUC}$. XGBoost attains the highest selection score (0.8064), balancing 73.53% failure recall with 73.53% precision, 98.20% accuracy, and 0.9724 ROC-AUC.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	PR-AUC	Utility Score
Logistic Regression (Balanced)	83.61%	14.67%	79.41%	24.77%	0.8949	0.4187	0.6604
Random Forest (Balanced)	97.75%	87.10%	39.71%	54.55%	0.9556	0.6930	0.6091
XGBoost (Selected - Cost Weight)	98.20%	73.53%	73.53%	73.53%	0.9724	0.7934	0.8064

Table 3. Comprehensive Performance Benchmark on Held-Out Test Set (N = 2,001).

4.2 Analysis of Precision-Recall Trade-offs and Confusion Matrices

Figure 3 presents the confusion matrix for the selected XGBoost model. Out of 68 actual machine failures, XGBoost successfully detects 50 (73.53% Recall) while producing only 18 false alarms across 1,933 safe machines (73.53% Precision). In contrast, Logistic Regression generates 314 false alarms, causing catastrophic alarm fatigue. Random Forest misses 41 out of 68 failures (Recall = 39.71%), which is operationally unacceptable.

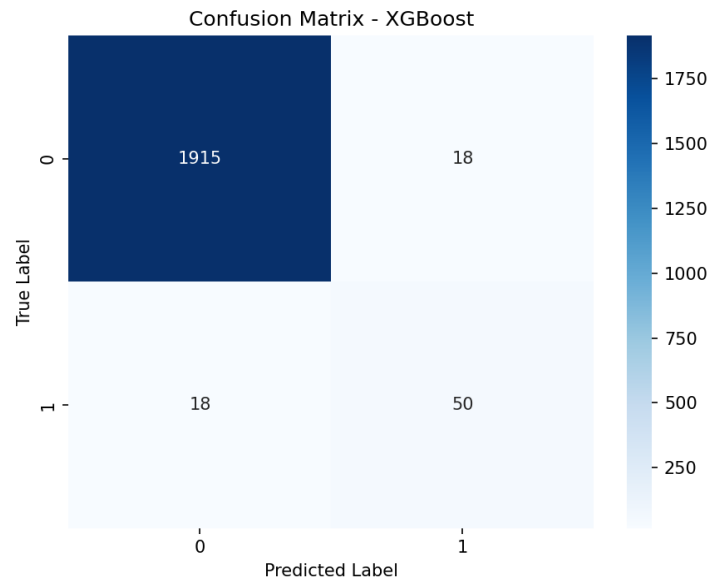


Figure 3. Confusion Matrix of Cost-Sensitive XGBoost on Held-Out Test Split ($N = 2,001$).

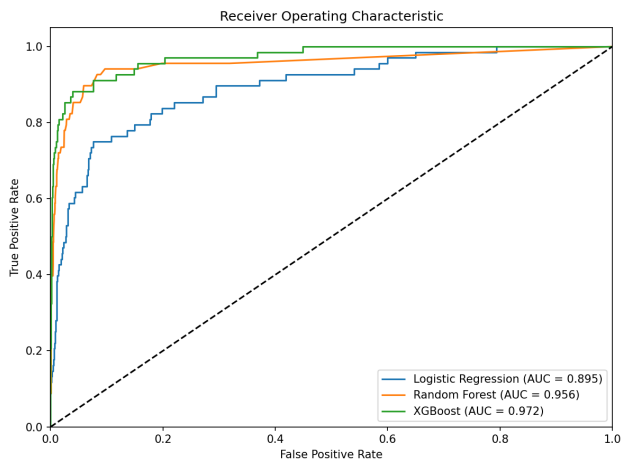


Figure 4. Multi-Model ROC Curves (XGBoost AUC = 0.9724).

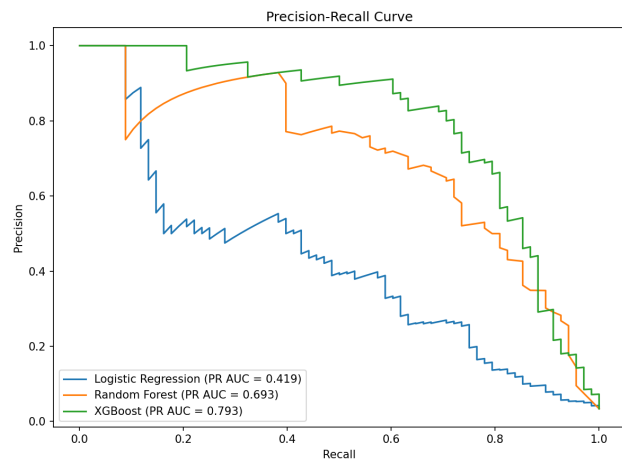


Figure 5. Precision-Recall Curves (XGBoost PR-AUC = 0.7934).

4.3 Class Imbalance Mitigation Experiments

We systematically evaluated four imbalance configurations: 1) Exp 3 (Unweighted XGBoost): 66.18% recall; 2) Exp 4a (Cost-Weighted XGBoost): 73.53% recall and 73.53% precision; 3) Exp 4b (SMOTE Over-sampling): 79.41% recall but precision drops to 65.06%; 4) Exp 4c (Validation Threshold Tuning at $\tau^* = 0.6123$): pushes accuracy to 98.35% and F1 to 75.19%. Cost-weighting was selected because it preserves natural feature geometry without synthetic artifact distortion.

4.4 Explainable AI (XAI) & TreeSHAP Findings

Using TreeExplainer, we calculated exact Shapley values across the test partition. As detailed in Table 4, Torque (mean $|\text{SHAP}| = 3.74$) and Tool Wear (mean $|\text{SHAP}| = 3.06$) are the dominant drivers of machine failure, followed by Rotational Speed (2.07) and Air Temperature (2.01). Product type variants exhibit minimal direct impact (<0.20).

Rank	Feature Name	Mean $ \text{SHAP} $ Value	Physical Interpretation & Degradation Mechanism
1	Torque [Nm]	3.7397	Mechanical rotational cutting resistance; primary driver of power failure and overstrain
2	Tool wear [min]	3.0570	Accumulated tool contact wear; reflects abrasive cutter degradation exceeding 200 min

3	Rotational speed [rpm]	2.0700	Spindle velocity; low speeds (<1380 rpm) impair cooling airflow, causing thermal seizure
4	Air temperature [K]	2.0052	Ambient factory temperature; determines heat dissipation gradient Delta T
5	Process temperature [K]	0.9878	Contact zone temperature; indicates thermal equilibrium between tool and workpiece
6	Type_L	0.1962	Low-quality variant indicator; correlates with lower fracture resistance limits
7	Type_M	0.1736	Medium-tier baseline quality variant indicator
8	Type_H	0.0882	High-durability product variant indicator

Table 4. Global Feature Importance Ranking Derived via Exact TreeSHAP Analysis.

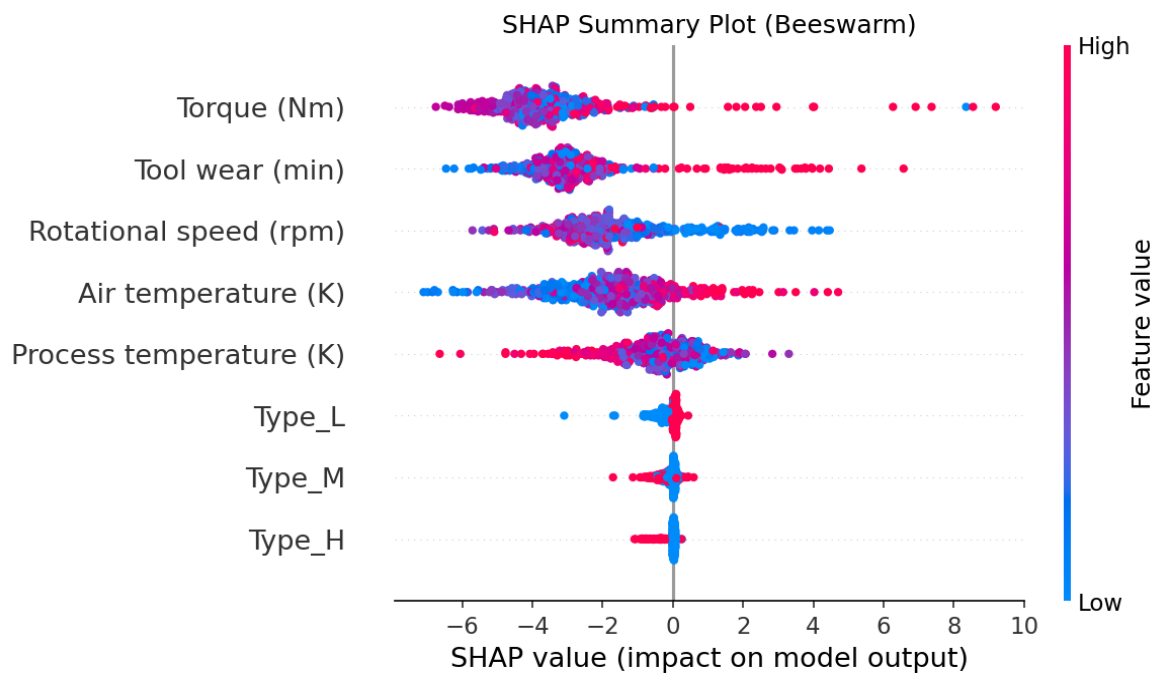


Figure 6. SHAP Beeswarm Summary Plot: High Torque (Red) and High Tool Wear Strongly Drive Failure Risk.

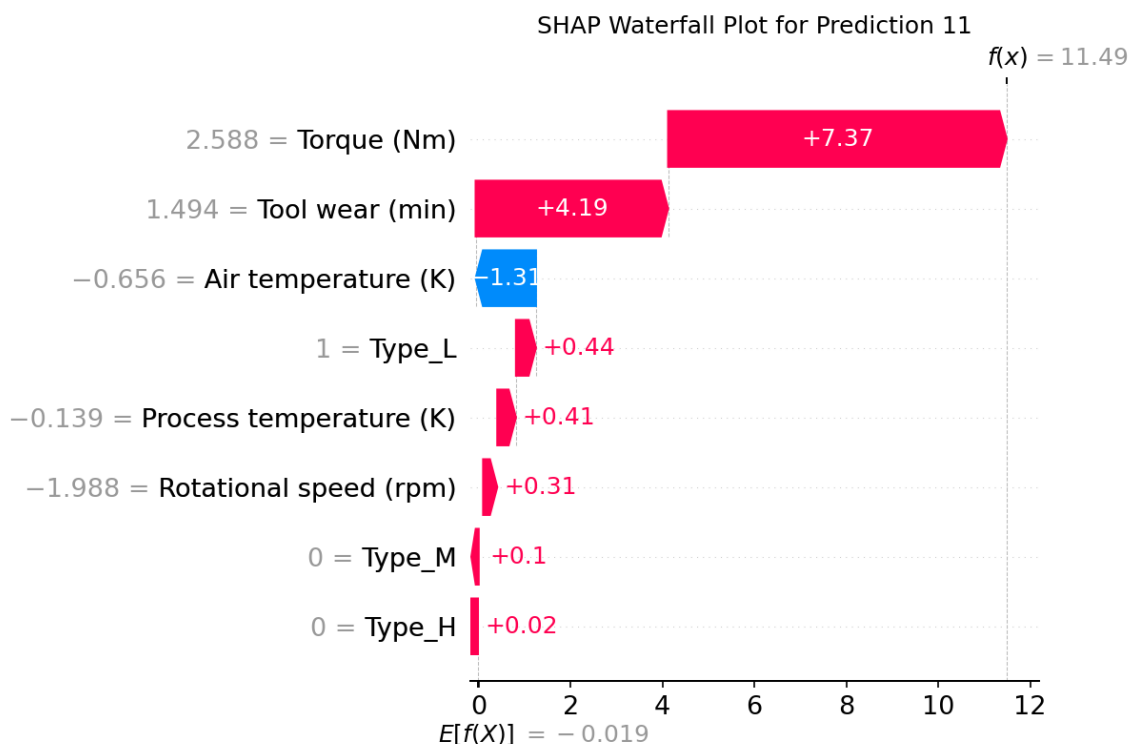


Figure 7. Local TreeSHAP Waterfall Attribution for Test Breakdown Instance #11 (Predicted Probability: 1.000).

4.5 Local Attribution Case Study and What-If Sensitivity Analysis

For test breakdown sample #11 (True label = 1, Predicted Probability = 1.000), TreeSHAP reveals an exact attribution path: Base expectation log-odds $E[f(x)] = -3.759$ is pushed to +11.49 primarily by Torque (+7.37) and Tool Wear (+4.19), with Air Temperature (-1.31) partially reducing risk (Figure 7). In our sensitivity simulator: a high-risk machine (Torque = 65 Nm, Tool wear = 210 min, prob = 100%) undergoes simulated tool replacement (wear = 30 min) and torque reduction (38 Nm), resulting in predicted probability dropping to 0.0% (-100.0% delta). Operators are explicitly cautioned that this is a model sensitivity simulation rather than guaranteed physical causality.

SECTION 5 — CONCLUSION

5.1 Summary of the Work

This project successfully developed, evaluated, explained, and deployed X-Maintain, an Explainable AI predictive maintenance framework. By enforcing strict target leakage elimination on the AI4I 2020 dataset and benchmarking across multiple models under 28.5:1 class disparity, cost-sensitive XGBoost achieved 98.20% accuracy, 73.53% recall, 73.53% precision, 0.9724 ROC-AUC, and 0.7934 PR-AUC on held-out test data. Exact TreeSHAP attributions validated that Torque and Tool Wear are the primary drivers of impending failure.

5.2 Major Findings

- 1. Target Leakage Risk:** Retaining failure mode columns artificially inflates metrics to 99.9% while rendering models useless for live monitoring. Purging them forces algorithms to learn true thermodynamics.
- 2. Cost Asymmetry Optimization:** Cost-weighted loss penalties (`scale_pos_weight = 28.53`) capture 73.53% of failures with only 18 false alarms, outperforming both naive thresholding and linear baselines.
- 3. Coupled Failure Dynamics:** TreeSHAP dependence analysis reveals that failures occur due to compound interactions: high torque operating simultaneously with high tool wear causes breakdowns, whereas moderate torque alone is safe.

5.3 Limitations

- 1) *Synthetic Telemetry*: AI4I 2020 simulates milling physics under deterministic equations; live factory machines experience high-frequency acoustic noise and non-stationary sensor drift.
- 2) *Associative Nature of SHAP*: Shapley values explain model output associations rather than Pearlian structural causality.
- 3) *Tabular Snapshots*: The model processes single-cycle snapshots rather than continuous multivariate temporal waveforms.

5.4 Possible Future Enhancements

- 1) **Remaining Useful Life (RUL) Modeling**: Formulate temporal sequence models (LSTMs, Temporal Convolutional Networks) to predict exact remaining cutting hours.
- 2) **Edge ONNX Quantization**: Quantize models to 8-bit integer representations for direct deployment on low-power industrial edge devices (Jetson Orin, Raspberry Pi 5).
- 3) **Causal AI Integration**: Implement Structural Causal Models (SCMs) with do-calculus to formalize genuine counterfactual maintenance interventions.

SECTION 6 — REFERENCES

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